once again, eternally regenerating their expanding three-dimensional worlds. To see how this goes, look at Figure 13.8, which illustrates one complete cycle, step by step.

At Stage 1, the two three-branes have just rushed toward each other and slammed together, and are now rebounding. The tremendous energy of the collision deposits a significant amount of high-temperature radiation and matter on each of the rebounding three-branes, and—this is key—Steinhardt and Turok argue that *the detailed properties of this matter and radiation have a nearly identical profile to what's produced in the inflationarymodel.* Although there is still some controversy on this point, Steinhardt and Turok therefore claim that the collision between the two three-branes results in physical conditions extremely close to what they'd be a moment after the burst of inflationary expansion in the more conventional approach discussed in Chapter 10. Not surprisingly, then, to a hypothetical observer within our three-brane, the next few stages in the cyclic cosmological model are essentially the same as those in the standard approach as illustrated in Figure 9.2 (where that figure is now interpreted as depicting evolution on one of the three-branes). Namely, as our three-brane rebounds from the collision, it expands and cools, and cosmic structures such as stars and galaxies gradually coalesce from the primordial plasma, as you can see in Stage 2. Then, inspired by the recent supernova observations discussed in Chapter 10, Steinhardt and Turok configure their model so that about 7 billion years into the cycle—Stage 3—the energy in ordinary matter and radiation becomes sufficiently diluted by the expansion of the brane so that a dark energy component gains the upper hand and, through its negative pressure, drives an era of accelerated expansion. (This requires an arbitrary tuning of details, but it allows the model to match observation, and so, the cyclic model's proponents argue, is well motivated.) About 7 billion years later, we humans find ourselves here on earth, at least in the current cycle, experiencing the early stages of the accelerated phase. Then, for roughly the next *trillion* years, not much new happens beyond our three-brane's continued accelerated expansion. That's long enough for our three-dimensional space to have stretched by such a colossal factor that matter and radiation are diluted almost completely away, leaving the braneworld looking almost completely empty and completely uniform: Stage 4.